

Lahore College for Women University
Department of Computer Science
Final Term Examination

Course Title: Data Structures and Algorithms
 Semester: III

Fall Term 2024
 Session: 2023 to 2027

Max Marks: 100

Total Time: 3hrs

Students Name: Roshan

Roll No: 095

Note: Cutting and rubbing is not allowed. Only use black and blue ink to solve this paper. Attach question papers in front of answer sheet.

Subjective (Total Marks: 40)

Note: Attempt all questions. All Questions carry equal marks.

Q#2 Given a 5 element queue Q (from front to back: 1, 3, 5, 7, 9), and an empty stack S, remove the elements one-by-one from Q and insert them into S, then remove them one-by-one from S and re-insert them into Q. The queue now looks like (from front to back). Explain each step. (10)

ind ↑

1 3 5 7 9

9
7
5
3
1

9
7
5
3
1

Q#2 What is closed hashing, how collisions will be resolved in it write an algorithm that insert N integer data values into a hash table and then searches a specific value from the hash table. Take an array of size N as data structure. (10)

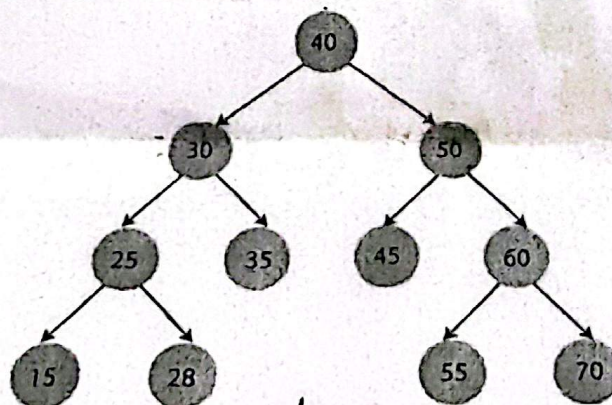
9 7 5 3 1

Q#3 Trace the process to sort the following array in ascending order using Quick sort algorithm. Also describe its algorithm and time complexity while ($U \neq P$ & $I \leq P$) swap(U & I) 5- (10)

5	7	1	2	10	4	9	3	6	8
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Q # 4 What are different traversing operation perform on Binary Search Trees. Apply inorder traversing on the following binary tree and write its sequence and algorithm using stacks as data structure. (10)

index = elem's no. of elem.



5 3 1 2 10 4 9 7 6 8

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